EXCEL FUNDAMENTALS & DATA MANAGEMENT

Consolidated learning notes | Lessons 1-8 | Concepts, menu paths, worked examples, expected results and key limitations

This reference consolidates the Excel topics studied from the supplied course transcripts and our guided practice. It is designed to follow directly in Excel; practice assignments and end-of-lesson quizzes have been omitted.

Contents

Lesson	Topics
1. Excel Fundamentals	Workbooks and worksheets; rows and cells; data types; copy, fill and navigation
2. Flash Fill	Joining, splitting, extracting, formatting, phone numbers, emails and limitations
3. Remove Duplicates	Single-column, multi-column and full-row duplicate removal
4. Text to Columns	Fixed Width, Delimited, destinations and limitations
5. Data Validation - Basics	Text Length, Whole Number, Decimal, Date, Time, dropdowns, messages
6. Custom Data Validation	Student IDs, current date, email structures and formulas
7. Worksheet Management	Insert, rename, delete, hide, unhide, move and copy
8. Cell Management	Column widths, row heights, AutoFit, Wrap Text and editing

Working convention: Excel menu paths refer to the desktop ribbon interface. Some menu wording, shortcuts, defaults and accepted date formats depend on the Excel edition, device and region.

LESSON 1 | Excel Fundamentals

Goal: navigate the spreadsheet interface, create a workbook, enter data and use essential filling tools.

1.1 Workbook, worksheet, row, column, cell and range

Term	Meaning	Example
Workbook	Excel file containing one or more worksheets	Excel_Complete_Practice.xlsx
Worksheet	Individual tab inside the workbook	01_Fundamentals
Column	Vertical cells identified by letters	A, B, C
Row	Horizontal cells identified by numbers	1, 2, 3
Cell	Intersection of a row and a column	B3
Range	Group of cells	A1:D6

1.2 Start, open and organize a workbook

1. On the Excel start page choose Blank workbook or open a recent, favorite or shared workbook.

2. To open an existing file use Open and browse to its folder. Templates and guided tours may also appear on the start page.

3. Save with Ctrl + S and give the file a meaningful name. Use the worksheet tabs to switch sheets; the + button adds one.

4. Use the Name Box near the Formula Bar to see the selected cell reference.

1.3 Enter data and recognize data types

Entry	Typical type	Default alignment
Rahul	Text	Left
100	Number	Right
01/01/2026	Date if recognized by Excel	Right
Employee101	Text	Left
250.50	Number	Right

> Important: Date interpretation depends on regional settings. Excel stores genuine dates as numeric serial values; the way they display is a format, not necessarily the original typing.

1.4 Copy, Paste, Fill Down and Fill Handle

1. Enter 2026 in H2. Copy with Ctrl + C and paste into H3 with Ctrl + V.

2. Enter 500 in H5, select H5:H10, then press Ctrl + D to fill down from the top cell.

3. Enter 1 in J2 and 2 in J3; select both and drag the small square at the bottom-right of the selection (the Fill Handle) down to create a series.

4. To duplicate an individual value, you can also drag the Fill Handle and choose the appropriate fill option.

1.5 Focus Cell

In supported Excel versions use View → Focus Cell. It highlights the active row and column, making large sheets easier to follow. Availability varies by version.

LESSON 2 | Flash Fill

Goal: give Excel an example of the desired output and let it infer a pattern across adjacent rows.

2.1 Where it is and how to use it

Menu: Home → Fill → Flash Fill. Windows shortcut: Ctrl + E.

1. Keep original data in neighboring columns and create a blank output column.

2. Enter one correct output example in the first row.

3. Select the next blank cell in the output column and press Ctrl + E.

4. Review all populated rows; for irregular records, supply more examples and repeat.

2.2 Join first and last names

A: First Name	B: Last Name	C: Full Name (expected)
Barry	Allen	Barry Allen
Darren	Hoffman	Darren Hoffman
Rosina	Smith	Rosina Smith

Type Barry Allen in C2; select C3 and press Ctrl + E. Flash Fill infers first name + space + last name.

2.3 Split a full name

A: Full Name	B: First Name	C: Last Name
Barry Allen	Barry	Allen
Darren Hoffman	Darren	Hoffman
Rosina Smith	Rosina	Smith

Type Barry in B2 → select B3 → Ctrl + E. Then type Allen in C2 → select C3 → Ctrl + E. Each output column is filled separately.

2.4 Extract date components

Date stored as text	Day	Month	Year
19/09/2018	19	09	2018
20/10/2019	20	10	2019
15/11/2020	15	11	2020

Give one example in each destination column and use Ctrl + E separately. For consistent behavior in this exercise, type the source date as text (prefix it with an apostrophe when entering). For real Excel dates, date functions are more reliable.

2.5 Capitalize names, extract initials and reformat numbers

Transformation	Original	Expected
Proper case	barry allen	Barry Allen
Uppercase	barry allen	BARRY ALLEN
Initials	Barry Allen	BA
Initials with punctuation	Barry Allen	B.A.
Phone formatting	919876543210	+91 98765 43210

Manually type the first desired output, select the row below and use Ctrl + E. Preserve source phone numbers as text to avoid losing leading zeros.

2.6 Extract numbers embedded in text

Source text	Output
Apple costs $45	45
TV costs USD 70	70
Banana costs 2.3 USD	2.3
Running took 1.5 hours	1.5
Product costs $4.25	4.25

One sample may be insufficient because numbers appear in different places and formats. Type the correct output for several different patterns (45, 70, 2.3, 1.5), then try Ctrl + E for the remaining rows.

2.7 Generate email addresses

First Name	Last Name	Expected Email
Barry	Allen	barry.allen@company.com
Darren	Hoffman	darren.hoffman@company.com
Rosina	Smith	rosina.smith@company.com

Type the first expected email in the output column, then apply Ctrl + E. Patterns may use dots, initials, fixed domain text or other characters.

2.8 Where Flash Fill fails

Situation	Explanation / better approach
09 → September	Mapping a number to a month name is not reliable; use TEXT(date,"mmmm") on a real date.
Currency appears before or after values	Mixed formatting often needs several examples and manual checking.
Only one name or optional middle names	First/last name patterns may break; inspect outliers.
Arithmetic	Flash Fill cannot calculate totals or margins; use formulas.
Source changed later	Flash Fill outputs are static values; rerun or use formulas if dynamic output is needed.
Wrong first example	The predicted result can be wrong; correct examples before repeating.

LESSON 3 | Remove Duplicates

Goal: remove repeated records using a business-defined key while keeping the first occurrence.

3.1 Core workflow

Menu: Data → Remove Duplicates.

1. Select the entire dataset, including the columns that must stay together.

2. Open Data → Remove Duplicates.

3. Check My data has headers when the first row contains field names.

4. Choose Unselect All and tick only the fields that define a duplicate, or choose Select All for full-row duplicates.

5. Click OK. Read Excel’s message showing how many duplicate values were removed and how many unique entries remain.

3.2 Single-column duplicate key

Customer ID	Customer Name	City
C101	Rahul	Bangalore
C102	Priya	Mumbai
C103	Amit	Delhi
C101	Rahul	Bangalore
C104	Sneha	Chennai
C102	Priya	Mumbai

Select the whole table; tick only Customer ID. Excel keeps the first C101 and C102 rows and removes their later repetitions: 6 rows become 4.

3.3 Whole-row duplicate check

Customer ID	Name	City
C101	Rahul	Bangalore
C102	Priya	Mumbai
C101	Rahul	Bangalore
C102	Priya	Pune

With all columns selected, only the repeated C101 / Rahul / Bangalore row is removed. The two C102 rows remain because Mumbai and Pune differ.

3.4 Composite key (multiple columns)

Customer ID	Order Date	Amount
C101	01-Jan-2026	500
C101	03-Jan-2026	900
C101	01-Jan-2026	500
C102	02-Jan-2026	700

Choose Customer ID + Order Date (not Amount). Excel removes the second occurrence of the C101 / 01-Jan-2026 combination.

3.5 Important safeguards

A repeated ID is not always an error: in transaction data the same customer can legitimately purchase multiple times. Choose the key according to business rules. Remove Duplicates changes the data and retains the first matching record; make a backup and verify the results. If headers are mistakenly unchecked, the first row can be treated as data.

LESSON 4 | Text to Columns

Goal: split one source field into two or more output columns using break positions or separators.

4.1 Two methods and the wizard

Menu: Data → Text to Columns. Windows ribbon shortcut: Alt, A, E (sequentially).

Method	How it works	Best use
Fixed Width	Splits at specified character positions	Same-length structured parts
Delimited	Splits wherever a selected separator appears	Names, dates and CSV-like strings

The wizard normally has three stages: choose method; set column breaks/delimiters and inspect the preview; then choose output column formats and the Destination.

4.2 Fixed Width: split the date text

Source text	1st segment	2nd segment	3rd segment
19/09/2018	19	/09	/2018
20/10/2019	20	/10	/2019

1. Enter dates as text: for instance, type '19/09/2018 in A2. Select every populated source row, not only A2.

2. Choose Fixed Width → Next. Put breaks after character 2 and character 5.

3. Choose Next; set a blank destination, e.g., B2, and finish.

4. Observe that the slash characters are retained as part of the segments. Double-click an unwanted break arrow in the wizard to remove it.

4.3 Fixed Width: phone numbers

Original (text)	Country Code	Mobile
919876543210	91	9876543210
919123456789	91	9123456789

Select the phone numbers → Fixed Width → break after two characters → destination in two empty columns → set output to Text if required → Finish. This works because the country code occupies the same positions in all records.

4.4 Delimited: full names

Full Name	Part 1	Part 2	Part 3
Barry Allen	Barry	Allen	
Darren Hoffman	Darren	Hoffman	
Tom Michael Hoffman	Tom	Michael	Hoffman

1. Select all populated full names. Choose Delimited → Next.

2. Tick Space only and check the preview; then click Next.

3. Set a blank destination (for example, J2) and finish.

4. Notice that delimiters disappear, and names with middle names may produce three or more output columns.

4.5 Delimited: dates and other separators

Source	Delimiter	Output columns
19/09/2018	/	19 | 09 | 2018
Bangalore,Karnataka,India	,	Bangalore | Karnataka | India
PRD-101	-	PRD | 101

In the Delimited wizard choose Other and enter the forward slash / for these dates (not backslash). To preserve 09 rather than converting it to 9, select Text for that output column at the final stage.

4.6 Limitations and overwrite warnings

Text to Columns does not reliably isolate numbers embedded in variable sentences (for example, “Apple costs $45” versus “Running took 1.5 hours”). Fixed Width struggles when character counts differ. A delimiter such as a space can produce too many columns, and overlapping the output with existing data can overwrite it after a warning. Always select the full input range and choose a clear destination area. Flash Fill may be preferable for inconsistent text.

LESSON 5 | Data Validation - Standard Rules

Goal: reduce entry errors using built-in validation criteria, clear instructions and error alerts.

5.1 Where to find the tool

Menu: Data → Data Validation → Data Validation. The dialog includes Settings, Input Message and Error Alert. The default Allow setting is Any Value, which imposes no restriction.

Allow option	Purpose
Whole Number	Integer restrictions
Decimal	Decimal/percentage restrictions
List	Limited set of choices
Date	Allowed date or date range
Time	Allowed time or time range
Text Length	Minimum, maximum or exact character count
Custom	Formula-based checks (Lesson 6)

5.2 Text Length - exactly 12

1. Select cell A2 → Data Validation → Settings.
2. Set Allow: Text Length, Data: Equal to, Length: 12.
3. Manually enter 1234: rejected; 123456789012: accepted; ABCD12345678: accepted as 12 characters.

> Important: Text Length does not verify that every character is a digit; use Custom for strict digit patterns. Do not store real passwords in an ordinary worksheet.

5.3 Text Length - between 10 and 16; maximum 20

Case	Allow	Data	Values	Examples
10-16 characters	Text Length	Between	Minimum 10, Maximum 16	ABC123 rejected; 10-16 accepted
Up to 20 characters	Text Length	Less than or equal to	Maximum 20	Excel accepted; 21+ rejected

5.4 Whole Number and Decimal

Scenario	Settings	Accepted	Rejected
Product Cost	Whole Number → Between → 100 and 240	100, 150, 240	99, 250, 150.5
Profit Margin	Decimal → Greater than or equal to → 0.35	35%, 40%	20%, 34%

Excel stores 35% as 0.35. Use Home → Number → Percentage to display 0.35 as 35%. For a margin limited to 35%-100%, use Decimal → Between → 0.35 and 1.

5.5 Date validation

Business rule	Settings	Example accepted / rejected
After 20-Dec-2025	Date → Greater than → 20-Dec-2025	21-Dec accepted; 20-Dec rejected
On/before 15-Jan-2026	Date → Less than or equal to → 15-Jan-2026	15-Jan accepted; 17-Jan rejected
Within 2025	Date → Between → 01-Jan-2025 and 31-Dec-2025	15-Jun-2025 accepted; Jan-2026 rejected

Dates must be recognized correctly under your regional settings. Enter =TODAY() in a cell to see today's date in that cell's current display format; inspect Excel and system settings for date order.

5.6 Time validation

Business rule	Settings	Example
Strictly after 11 AM	Time → Greater than → 11:00:00 AM	11:00 rejected; 11:15 accepted
Between 11 AM and 10 PM	Time → Between → 11:00:00 AM and 10:00:00 PM	11:00 and 22:00 accepted; 22:30 rejected

> Important: Use a colon (:) in times, not a dot. "Between" includes both boundaries; "Greater than" excludes the stated value.

5.7 Input Message and Error Alert

Tab / setting	Purpose	Example
Input Message	Shown when the validated cell is selected	Title: Date Input; message: Enter a date after Dec 20, 2025.
Error Alert: Stop	Rejects invalid manual entry	Title: Invalid Date; message: Enter a valid date.
Error Alert: Warning	Warns but can be overridden	Useful when review, not strict enforcement, is intended
Error Alert: Information	Informational message; invalid entry can be accepted	Nonblocking guidance

1. Select a validated cell → reopen Data Validation.
2. On Input Message, tick the checkbox and enter title and instructions.
3. On Error Alert, enable the alert, choose Stop, and add a clear title and correction text.
4. Click OK; select the cell to see instructions, then enter an invalid value to see the Stop message.

5.8 List validation: typed choices

1. Select a cell (e.g., K2). Choose Allow: List.
2. In Source type Pizza,Burger,Fries (the list separator can depend on region).
3. Leave In-cell dropdown checked; enable Stop for invalid items.
4. The dropdown shows Pizza, Burger and Fries. Users may type an allowed item manually.

5.9 List validation: use source cells

Cell	Source value
M2	French Fries
M3	Sandwich
M4	Hamburger
M5	Bread Toast
M6	Mojito (added later)
M7	Blank reserved

For a dropdown in L2, choose List and set the Source to =$M$2:$M$7. Adding Mojito in M6 updates the choices because M6 lies inside the referenced range. Adding a value in M8 will not automatically extend this fixed range.

5.10 Copy validation to other rows

1. Copy the configured cell with Ctrl + C.

2. Select the destination range (e.g., A3:A20).

3. Choose Home → Paste → Paste Special → Validation to copy only the rule. A normal paste may also copy the cell content/format.

4. Alternatively select all target cells before configuring the rule.

> Important: Data Validation is a strong aid for manual data entry, not a full data-integrity guarantee: paste/import operations may bypass rules, and existing invalid values may remain.

5.11 Product-entry example from the course

Column	Rule / setting
Product ID	Text Length = 6
Product Category	List: Pizza, Burger, Fries
Product Cost	Whole Number between 100 and 240
Product Margin	Decimal greater than or equal to 0.35 (35%)
Product Sales Date	Date between 01-Jan-2025 and 31-Dec-2025

Configure the five rules in the first data row, add appropriate input/error messages, then copy only the validation rules down the relevant columns. The course solution briefly mentions 2023 for the sale-date bound; the stated assignment rule was within 2025, which is retained here.

LESSON 6 | Custom Data Validation

Goal: use formulas to enforce rules that are too specific for a single built-in validation type.

6.1 Basic idea and application

Select the target cell → Data → Data Validation → Settings → Allow: Custom → enter a formula beginning with =. The entry is permitted when the expression evaluates to TRUE. Set an Error Alert (Stop) to reject invalid manual inputs.

6.2 Student ID with fixed prefix and four digits

Course scenario: a seven-character student ID. The narration is inconsistent about the prefix (it mentions S2 but demonstrates STU). For the worked example below, the format is explicitly STU + four digits, such as STU3256.

```
=AND(LEN(F7)=7,EXACT(LEFT(F7,3),"STU"),IFERROR(RIGHT(F7,4)=TEXT(VALUE(RIGHT(F7,4)),"0000"),FALSE))
```

Input	Expected
STU3256	Accepted
STU9876	Accepted
STU123	Rejected: too short
ABC1234	Rejected: wrong prefix
STU12A4	Rejected: non-digit
STU12345	Rejected: too long

1. Put the Student ID header in F6 and select F7.
2. Set Allow → Custom and paste the formula above.
3. Set Error Alert → Stop → “Enter STU followed by exactly four digits.”
4. Test each sample above manually. Copy validation down with Paste Special → Validation.

6.3 Functions appearing in the student ID rule

Function	What it does	Example
LEN	Count characters	LEN("STU1234") → 7
LEFT	Take leading text	LEFT("STU1234",3) → STU
RIGHT	Take trailing text	RIGHT("STU1234",4) → 1234
EXACT	Compare case sensitively	EXACT("STU","stu") → FALSE
AND	Require all tests to pass	AND(TRUE,FALSE) → FALSE
VALUE	Convert numeric text to number	VALUE("123") → 123
TEXT	Format numeric value as text	TEXT(35,"0000") → 0035
IFERROR	Handle failed expressions	IFERROR(1/0,FALSE) → FALSE

6.4 Accept today’s date only

For a daily tracking field in G7:

```
=AND(ISNUMBER(G7),G7=TODAY())
```

Set Data Validation → Custom; paste the formula; configure the Stop alert “Only today’s date is allowed.” The formula compares an entered numeric Excel date with the current computer date. Date display formatting can vary; the underlying date must match.

> Important: Already-entered dates do not automatically change or disappear the next day. The validation is generally evaluated on new manual entry/edit, not continuously.

6.5 Basic email-address structure

For a customer email in H7, the course highlights missing @ signs, incomplete domains, multiple @ signs and unwanted spaces. A simple structural check is:

```
=AND(ISERROR(FIND(" ",H7)),LEN(H7)-LEN(SUBSTITUTE(H7,"@",""))=1,IFERROR(FIND("@",H7)>1,FALSE),IFERROR(FIND(".",H7,FIND("@",H7)+2)<LEN(H7),FALSE))
```

Input	Expected
customer@gmail.com	Accepted
employee@yahoo.com	Accepted
customer.com	Rejected: no @
customer@gmail	Rejected: missing dot in domain
customer @gmail.com	Rejected: contains a space
customer@@gmail.com	Rejected: multiple @

Select H7 → Custom → paste the formula → Stop alert. Test by manually typing into the validated cell. This is a basic pattern check, not verification that an inbox exists or exhaustive compliance with every valid email format.

6.6 Functions useful for email checks

Function	Example	Meaning
FIND	FIND("@","customer@gmail.com") → 9	Locate a character
SUBSTITUTE	SUBSTITUTE("A-B","-","") → AB	Replace or remove text
ISERROR	ISERROR(FIND(" ",H7))	Detect lookup errors
ISNUMBER	ISNUMBER(G7)	Check that the stored value is numeric

6.7 Check whether prerequisite fields are filled

To allow H7 only after the Student ID in F7 and date in G7 are populated, an example condition is:

```
=AND(F7<>"",G7<>"",H7<>"")
```

This replaces, rather than adds to, the email-format rule if pasted alone into H7. To enforce both requirements, combine both sets of conditions inside one AND expression. Field-presence checks do not independently validate the correctness of those fields.

6.8 Generating custom validation with ChatGPT

1. Describe the specific business rule and the exact target cell (for example F7).

2. Specify all accepted and rejected patterns, including edge cases such as spaces, leading zeros and missing fields.

3. Ask for an Excel Custom Data Validation formula and an explanation.

4. Paste it under Allow → Custom, set Stop, then test with both valid and invalid manual entries.

5. Use Paste Special → Validation to copy rules to later rows, checking relative references.

> Important: AI-generated formulas can contain errors or omit requirements. Verify them before applying rules to production data.

LESSON 7 | Worksheet Management

Goal: organize, rename, hide, duplicate and transfer sheets while preserving workbook integrity.

7.1 Workbook and worksheet basics

A workbook may contain multiple worksheets for source data, customer details, analysis and dashboards. Modern Excel commonly starts with one sheet, but settings and older versions vary.

7.2 Insert a new sheet

1. Select the worksheet before which you want a new sheet.

2. Choose Home → Insert → Insert Sheet; the new sheet appears before the current one.

3. Alternatively click the + control at the sheet tabs (typically inserts at the end) or press Shift + F11.

7.3 Rename and naming restrictions

1. Double-click a worksheet tab, or select Home → Format → Rename Sheet.

2. Type a descriptive name such as Sales Data and press Enter.

Names may include spaces but are limited to 31 characters. Names must be unique within the workbook; : \ / ? * [ ] are not allowed.

7.4 Delete and recoverability

Right-click the worksheet tab → Delete, or Home → Delete → Delete Sheet. If Excel warns that the deletion is permanent, review carefully before confirming. Deleted worksheets generally cannot be restored with Ctrl + Z. Retain a backup or earlier file version.

7.5 Hide and unhide

1. Right-click Sales Data → Hide. The tab disappears, but data and underlying workbook content remain.

2. Right-click another worksheet tab → Unhide → choose Sales Data → OK.

> Important: Hide is for convenience, not for confidentiality. Another recipient can often unhide the sheet or inspect its data by other means.

7.6 Move or copy within a workbook

1. Right-click a sheet tab → Move or Copy.

2. Choose the current workbook in To book and the destination in Before sheet.

3. Leave Create a copy unchecked to move; check it to duplicate.

4. You may also drag a sheet tab to reorder it; Ctrl + drag commonly creates a copy on Windows desktop Excel.

7.7 Copy a dashboard into a separate workbook

1. Prepare a workbook with Sales Data and Dashboard worksheets.

2. Right-click Dashboard → Move or Copy.

3. Under To book select (new book); check Create a copy.

4. Click OK; save the new workbook, e.g., Sales_Dashboard.xlsx.

5. Inspect copied charts and formulas: they may still reference the original workbook if their underlying data was not copied.

Action	Original remains?	Result
Move to another workbook (no copy)	No	Worksheet moves to target file
Copy to another workbook	Yes	Duplicate created in target file
Copy within same workbook	Yes	Duplicate tab created

7.8 Why copy a sheet instead of pasting only a chart?

Moving/copying the full worksheet usually preserves more of its formatting, column widths, row heights, gridline settings, page layout and chart arrangement. Copying only a chart may leave those worksheet-level choices behind.

LESSON 8 | Cell Management

Goal: make data legible by controlling column widths, row heights, wrapping and editing.

8.1 Width and height are formatting properties

A column controls horizontal room for its cells and a row controls vertical room. The lecture's example uses width 10 and height 16; defaults can vary. Right-click a column letter → Column Width, or a row number → Row Height to inspect or change it.

8.2 Worked customer dataset

Customer ID	Name	Email	Address
C101	Jessica Walsh	jessica.walsh@gmail.com	123 MG Road, Bangalore, Karnataka
C102	Christine Anderson	christine.anderson@hotmail.com	45 Park Street, Kolkata, West Bengal
C103	Rahul Sharma	rahul.sharma@gmail.com	78 Anna Salai, Chennai, Tamil Nadu

8.3 AutoFit one column or all columns

1. Select the Customer Name column B.

2. Choose Home → Format → AutoFit Column Width. Excel fits the longest displayed name in the selected cells.

3. Alternatively double-click the boundary between headers B and C.

4. To adjust the whole dataset, click inside it and press Ctrl + A (confirm selection) → Home → Format → AutoFit Column Width.

5. Notice that email and address columns may become very wide and require horizontal scrolling.

8.4 AutoFit row height

1. Make row 3 deliberately tall using right-click row 3 → Row Height → 60.

2. Select row 3 → Home → Format → AutoFit Row Height (or double-click the row boundary).

3. Excel resizes the row according to the displayed contents. Long wrapped text is a common reason to use this feature.

8.5 Wrap Text: keep a narrow address column

1. Select the Address column (e.g., E2:E6).

2. Choose Home → Wrap Text.

3. Set column width to 20 using right-click on the column header → Column Width.

4. Select the affected rows → Home → Format → AutoFit Row Height.

5. The address stays in one cell, but appears on multiple visual lines; row heights may now differ.

Tool	Changes	Best use
AutoFit Column Width	Column width based on contents	Names, emails, one-line values

Tool	Changes	Best use
AutoFit Row Height	Row height based on contents	Wrapped descriptions
Wrap Text	Line breaks within a displayed cell	Addresses, notes and remarks
Manual width/height	Exact user-controlled dimension	Consistent dashboard layouts

8.6 Resize manually

Drag the boundary between two column letters to change column width, or between two row numbers to change row height. Alternatively enter an explicit value in Column Width or Row Height.

8.7 Edit an existing cell without replacing it

1. Select the cell containing James94@hotmail.com.

2. Double-click it or press F2 to enter edit mode.

3. Place the cursor after “James”, type an underscore and press Enter.

4. The cell now reads James_94@hotmail.com. You can also edit the text in the Formula Bar.

Starting to type on a selected cell without entering edit mode replaces the existing content. F2 or double-click lets you modify just the required characters.

8.8 Quick shortcut reference

Action	Shortcut / method
Copy / Paste / Undo	Ctrl + C / Ctrl + V / Ctrl + Z
Edit cell	F2 or double-click cell
Select current contiguous data	Ctrl + A inside the dataset
AutoFit column	Double-click right edge of column header
AutoFit row	Double-click lower edge of row header
Save workbook	Ctrl + S
Flash Fill	Ctrl + E
Insert worksheet	Shift + F11
Fill down	Ctrl + D

END OF LESSONS 1-8

Included: explanations, step-by-step operations, sample inputs and outputs, formulas introduced in the lessons, and practical warnings. Excluded as requested: practice assignments and quizzes.